

Diffusion Models

Suppose data x_0 follows some unknown distribution $q(x_0)$.

Similar to VAE, in diffusion models, we have the latent variables $x_{1:T}$ with the assumption that $x_{0:T}$ form a Markov chain with Gaussian transitions, starting with $x_T \sim \mathcal{N}(0, I)$,

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

for $1 \leq t \leq T$ and the likelihood

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t).$$

This corresponds to the reverse diffusion process described later.

Unlike VAE where we approximate the posterior, the posteriors in diffusion models are assumed to follow a Markov chain:

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}).$$

The conditional probabilities $q(x_t|x_{t-1})$ are Gaussians:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I),$$

where $0 < \beta_t < 1$ are constants. The posteriors correspond to the forward diffusion process.

Objective Function

Similar to VAE, taking $x_{1:T}$ as the latent variables, the ELBO is

$$\mathbb{E}_{q(x_0)} [-\log(p_\theta(x_0))] \leq \mathbb{E}_{q(x_{0:T})} \left[\log \frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})} \right].$$

Since

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1})$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t),$$

the ELBO expands

$$\begin{aligned} & \mathbb{E}_{q(x_{0:T})} \left[\log \frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})} \right] \\ &= \mathbb{E}_{q(x_{0:T})} \left[-\log p(x_T) + \sum_{t=2}^T \log \frac{q(x_t|x_{t-1})}{p_\theta(x_{t-1}|x_t)} + \log \frac{q(x_1|x_0)}{p_\theta(x_0|x_1)} \right] \\ &= \mathbb{E}_{q(x_{0:T})} \left[-\log p(x_T) + \sum_{t=2}^T \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \cdot \frac{q(x_t|x_0)}{q(x_{t-1}|x_0)} + \log \frac{q(x_1|x_0)}{p_\theta(x_0|x_1)} \right] \\ &= \mathbb{E}_{q(x_{0:T})} \left[\log \frac{q(x_T|x_0)}{p(x_T)} + \sum_{t=2}^T \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} - \log p_\theta(x_0|x_1) \right] \\ &= \mathbb{E}_{q(x_0)} [\text{KL}(q(x_T|x_0) \parallel p(x_T))] \\ & \quad + \mathbb{E}_{q(x_{0:T})} \left[\sum_{t=2}^T \text{KL}(q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t)) \right] \\ & \quad - \mathbb{E}_{q(x_0, x_1)} \log p_\theta(x_0|x_1) \\ &:= L_T + \sum_{t=2}^T L_{t-1} + L_0. \end{aligned}$$

We will revisit the ELBO and see why we further condition on x_0 in above equations.

Forward Diffusion Process

By reparametrizing, for independent $\varepsilon_i \sim \mathcal{N}(0, I)$ and $\alpha_t = 1 - \beta_t$,

$$\begin{aligned} x_t &= \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_{t-1} \\ &= \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_{t-1}. \end{aligned}$$

Also,

$$x_{t-1} = \sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} \varepsilon_{t-2}.$$

Hence,

$$x_t = \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \varepsilon_{t-2} + \sqrt{1 - \alpha_t} \varepsilon_{t-1}.$$

Since ε_{t-2} and ε_{t-1} are independent,

$$\sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \varepsilon_{t-2} + \sqrt{1 - \alpha_t} \varepsilon_{t-1} \sim \mathcal{N}(0, (1 - \alpha_t \alpha_{t-1})I).$$

We reparametrize

$$x_t = \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \tilde{\varepsilon}_{t-2}$$

for $\tilde{\varepsilon}_{t-2} \sim \mathcal{N}(0, I)$.

Continue this way and we have

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon \quad (*)$$

for $\varepsilon \sim \mathcal{N}(0, I)$ and $\bar{\alpha}_t = \alpha_1 \dots \alpha_t$.

In other words

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, \sqrt{1 - \bar{\alpha}_t} I).$$

Notice that if $\bar{\alpha}_t \rightarrow 0$, then $q(x_t | x_0) \rightarrow \mathcal{N}(0, I)$. This is desired as we want pure Gaussian noise after the forward diffusion process.

Reverse Diffusion Process

Notice that the conditional probability $q(x_{t-1}|x_t)$ is tractable when conditioning further on x_0 using Bayes' rule:

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)} \\ &= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)} \quad (\text{Markov Property}) \end{aligned}$$

Since

$$\begin{aligned} q(x_t|x_{t-1}) &= \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t I) \\ q(x_{t-1}|x_0) &= \mathcal{N}(x_{t-1}; \sqrt{\bar{\alpha}_{t-1}} x_0, \sqrt{1 - \bar{\alpha}_{t-1}} I) \\ q(x_t|x_0) &= \mathcal{N}(x_t; \sqrt{\alpha_t} x_0, \sqrt{1 - \alpha_t} I), \end{aligned}$$

we have

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t I)$$

for

$$\begin{aligned} \tilde{\mu}_t &= \tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} x_0 \\ \tilde{\beta}_t &= \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} (1 - \alpha_t) \end{aligned}$$

with some constant terms omitted.

Because of the parametrization (*),

$$\tilde{\mu}_t = \tilde{\mu}_t(x_t, \varepsilon) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon \right)$$

for $\varepsilon \sim \mathcal{N}(0, I)$.

Objective Function Revisited

Note that we have nothing to optimize in the L_T term and therefore L_T is omitted from the ELBO.

For L_t with $1 \leq t < T$, it is the KL-divergence between two Gaussian distributions:

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \mathcal{N}(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t I) \\ p_\theta(x_{t-1}|x_t) &= \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)). \end{aligned}$$

Hence, we set $\Sigma_\theta(x_t, t) = \sigma_t^2 I := \tilde{\beta}_t I$ (without learnable parameters).

$\mu_\theta(x_t, t)$ should predict

$$\tilde{\mu}_t = \tilde{\mu}_t(x_t, \varepsilon) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon \right)$$

for some $\varepsilon \sim \mathcal{N}(0, I)$. Therefore we design

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right)$$

for some neural network $\varepsilon_\theta(x_t, t)$ predicting ε . The objective function L_{t-1} is

$$\begin{aligned} L_{t-1} &= \mathbb{E}_{q(x_0, x_{t-1}, x_t)} [\text{KL}(q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t))] \\ &= \mathbb{E}_{x_0, \varepsilon} \left[\frac{1}{2\sigma_t^2} \|\mu_\theta(x_t(x_0, \varepsilon), t) - \tilde{\mu}_t(x_t(x_0, \varepsilon), \varepsilon)\|^2 \right] \\ &= \mathbb{E}_{x_0, \varepsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\varepsilon - \varepsilon_\theta(x_t(x_0, \varepsilon), t)\|^2 \right], \end{aligned}$$

where $x_t(x_0, \varepsilon) = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$.

L_0 will be dealt separately according to $p_\theta(x_0|x_1) = \mathcal{N}(x_0; \mu_\theta(x_1, 1), \sigma_1^2 I)$.

Simplified Objective Function

The objective function $L(\theta) = \sum_{t=2}^T L_{t-1}(\theta) + L_0(\theta)$ can be simplified to

$$L_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \varepsilon, t} [\|\varepsilon - \varepsilon_\theta(x_t(x_0, \varepsilon), t)\|^2].$$

The paper proposed following training and sampling algorithms:

Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
      $\nabla_{\theta} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$   
6: until converged
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Algorithm 2 Sampling

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1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
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